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GAUSSIAN CLASS-REPRESENTATION CAN OVERCOME CATASTROPHIC FORGETTING IN OBJECT CLASSIFICATION

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CONTINUAL LEARNING FOR OBJECT CLASSIFICATION USING GAUSSIAN MODELING (PRELIMINARY TITLE)

Esta investigación aborda el desafío del aprendizaje continuo (continual learning) para la clasificación y detección de objetos en tiempo real. Se centra en integrar un extractor de características preentrenado de forma no supervisada con modelos Gaussianos independientes para caracterizar las clases de objetos.

El objetivo principal de este trabajo es desarrollar y validar un sistema capaz de incorporar nuevas clases de manera incremental sin reentrenar todo el modelo, mitigando el olvido catastrófico de categorías aprendidas previamente y permitiendo la detección automática de clases desconocidas en escenarios de aplicación real.

Para lograr esto, el estudio evalúa la efectividad de representaciones Gaussianas separables y no separables, analiza su compacidad y eficiencia de muestras, y explora su extensión a pipelines de detección de objetos reemplazando los cabezales de clasificación tradicionales por modelos Gaussianos. El enfoque propuesto busca combinar la independencia de los modelos Gaussianos con la capacidad de generalización del aprendizaje de características no supervisado, proporcionando una solución escalable y eficiente para entornos dinámicos donde emergen continuamente nuevas clases de objetos.

CONTINUAL LEARNING FOR OBJECT CLASSIFICATION USING GAUSSIAN MODELING (PRELIMINARY TITLE)

This research addresses the challenge of continual learning for real-time object classification and detection. It focuses on integrating an unsupervised pretrained feature extractor with independent Gaussian models to characterize object classes. The main objective of this work is to develop and validate a system capable of incrementally incorporating new classes without retraining the entire model, mitigating catastrophic forgetting of previously learned categories, and enabling the automatic detection of unknown classes in real application scenarios. To achieve this, the study evaluates the effectiveness of separable and non-separable Gaussian representations, analyzes their compactness and sample efficiency, and explores their extension to object detection pipelines by replacing traditional classification heads with Gaussian models. The proposed approach seeks to combine the independence of Gaussian models with the generalization capacity of unsupervised feature learning, providing a scalable and efficient solution for dynamic environments where new object classes continuously emerge.

Abstract

Los modelos convencionales de aprendizaje profundo (Deep Learning) suelen ser monolíticos y tienen dificultades para adaptarse cuando se exponen a nuevas clases, requiriendo típicamente un reentrenamiento con el conjunto de datos completo. Esto limita su aplicabilidad en sistemas del mundo real que deben evolucionar e incorporar información nueva con el tiempo. Para abordar estos desafíos, este trabajo introduce un enfoque de aprendizaje continuo (Continual Learning) que desacopla la extracción de características de la clasificación, combinando un extractor de características preentrenado de forma no supervisada con modelos Gaussianos independientes que capturan la representación de cada clase. Además, extendemos el método a la detección de objetos en tiempo real reemplazando los cabezales de clasificación estándar con nuestros modelos basados en Gaussianas y exploramos el uso de autoencoders como un extractor de características alternativo optimizado para CL. Adicionalmente, abordamos el problema crítico de identificar automáticamente cuándo debe introducirse una nueva clase. Los experimentos en el benchmark a gran escala ImageNet y en otros conjuntos de datos como Caltech-256 y CIFAR-100 revelan que nuestro método alcanza un rendimiento a la par con los enfoques del estado del arte basados en prompts, mientras se añaden nuevas clases secuencialmente. Estos resultados demuestran que el enfoque propuesto no solo es efectivo y escalable, sino también más ligero y sencillo de implementar en comparación con los enfoques existentes. Esta simplicidad facilita un aprendizaje clase por clase fluido, permitiendo al mismo tiempo una adaptación continua a entornos del mundo real dinámicos y en evolución.

Abstract

Conventional deep learning models are often monolithic and struggle to adapt when exposed to new classes, typically requiring retraining on the entire dataset. This limits their applicability in real-world systems that must evolve and incorporate novel information over time. To address these challenges, this work introduces a continual learning (CL) approach that decouples feature extraction from classification, pairing an unsupervised pretrained feature extractor with independent Gaussian models that capture each class representation. We further extend the method to real-time object detection by replacing standard classification heads with our Gaussian-based models and explore autoencoders as an alternative feature extractor optimized for CL. Additionally, we tackle the critical problem of automatically identifying when a new class should be introduced. Experiments on the large-scale ImageNet benchmark and other datasets like Caltech-256 and CIFAR-100 reveal that our method reaches performance on par with prompt-based state-of-the-art approaches, while new classes are sequentially added. These results demonstrate that the proposed approach is not only effective and scalable, but also lighter and simpler to implement compared to the existing approaches. This simplicity facilitates seamless class-by-class learning while enabling continuous adaptation to dynamic and evolving real-world environments.

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Chapter 1

Introduction

1.1 Motivation

State-of-the-art object recognition methods utilize deep learning models that are either trained end-to-end on the target application or initially trained in one application and then fine-tuned for a different target application. In either case, they are monolithic models: the same single model is normally trained on a vast dataset and used for feature extraction and classification. These models are highly effective when the set of object classes to be recognized is fixed and known at the training/fine-tuning time. However, when new object classes need to be incorporated, the model must be retrained or fine-tuned. This training requires significant computational resources and access to the entire training dataset, and in some cases leads to catastrophic forgetting [1].

One way to integrate new classes while avoiding catastrophic forgetting, is to approach the problem using continual or class-incremental object recognition, where an algorithm must learn to distinguish between a growing number of object classes [2]. In the context of continual learning, a task refers to a distinct portion of the training experience, usually presented sequentially to the model. Instead of having access to the entire dataset at once, as in standard supervised learning, the model receives data in sequential chunks [3]. Each task generally correspond to a set of new classes to recognize. The main challenge is that when the model is trained task by task, it tends to overwrite previously learned knowledge. Given that current monolithic models are not capable of handling this type of application, an alternative paradigm is required.

This limitation has motivated SOTA methods for object classification to focus on balancing stability and plasticity. Stability refers to a model's ability to preserve previously learned knowledge, while plasticity describes its capacity to adapt and incorporate new tasks [4]. However, traditional approaches rarely optimize both aspects simultaneously: when stability dominates, the model struggles to integrate new knowledge; when plasticity dominates, it loses access to past information. This trade-off is compounded by the limited capacity of neural networks to integrate new representations without interfering with old ones, and by the need to continually adjust the network's structure to match the increasing number of

object classes.

One solution to this problem is to divide the monolithic models into two distinct components: one responsible for feature extraction and the other for object classification. These components must be trained separately, at different times. The feature extraction model must be trained when the recognition system is initialized/created, and the obtained features must be general enough to characterize all possible classes of objects to be seen (not just the ones known at training time). The object classification model should be easily updated or trained as new object classes are detected. Ideally, it should be able to incorporate new object classes without modifying the parts of the model responsible for recognizing the already known object classes. This mirrors the functionality of the human visual system, where the structures responsible for the extraction of characteristics (simple and complex cells in cortical areas V1 and V2) undergo adaptation during the initial postnatal weeks/months [5]. However, the ability to learn new categories of objects persists throughout life. Object recognition occurs in high-level cognitive areas, and acquiring knowledge of a new object class can result in a slight reduction in the ability to recognize similar classes of objects over time.

We are inspired by this idea and propose a continual learning object recognition system based on two components. The first component is a feature extraction model built using a pretrained foundation model. The second component is a statistical classifier built using various Gaussian models. Each model (single Gaussian or mixture of Gaussians) characterizes a specific object class independently of the others. Since these Gaussian models can be trained or built at different times, a new Gaussian model is created to characterize a new object class when it needs to be incorporated. For classification, the class whose Gaussian model yields the highest likelihood of the input features is selected. This approach is analogous to a Bayes classifier, in which new classes are incorporated over time.

Gaussian models offer two key advantages for characterizing object classes. First, each object class model operates independently from the others, facilitating the straightforward incorporation of new classes into the model. Second, Gaussian models provide a compact representation of object classes, requiring few samples to construct a model that can be refined as new samples become available.

1.2 Hypothesis

We hypothesize that the use of independent Gaussian models, combined with an unsupervised pretrained feature extractor, enables continual learning that can incorporate new classes in object classification and detection without retraining the entire model. This design mitigates catastrophic forgetting and further allows the automatic detection of previously unknown classes.

1.3 General Objectives

The main objective of this work is to develop and validate a continual learning framework for real-time object classification and detection, based on an unsupervised pre-trained feature extractor and independent Gaussian models, with the purpose of enabling the incremental incorporation of new classes, preventing catastrophic forgetting of previously learned classes, and enabling the automatic detection of unknown classes in real application scenarios.

1.4 Specific Objectives

For the purpose of fulfilling the general objective and verifying the hypothesis of this work, the following specific objectives are defined:

- Evaluate Gaussian representations in continual learning. Empirically verify that representing classes using separable and non-separable Gaussians enables stable continual learning without significant degradation in previously learned classes.
- Explore autoencoders as feature extractors. Research, train, and validate an autoencoder that achieves performance comparable to the backbones used so far as feature extractors, but optimized for continual learning environments.
- Extend the approach to real-time object detection. Investigate and apply object detection architectures in which the classification head can be modified and replaced by Gaussian models to allow the incremental incorporation of new classes.
- Implement automatic detection of new classes. Develop a mechanism to automatically identify when an observed object does not belong to any known class, in order to dynamically create new classes in the system.

Chapter 2

Theoretical framework

In this chapter we will present the necessary concepts to fully understand the work done. First, we will review general concepts of artificial intelligence, Machine Learning, Image Processing, Object Classification and Object Detection. Second, we will describe the datasets used in this work and finally we will talk about Continual Learning and describe some of the state-of-the-art methods.

2.1 Artificial Intelligence

Artificial Intelligence is a broad field of computer science dedicated to creating systems that can perform tasks that typically require human intelligence, such as learning, reasoning, problem-solving, perception, and language understanding [6].

2.1.1 Machine Learning

Machine Learning is a subfield of AI focused on the development of algorithms that allow computers to learn from data without being explicitly programmed. ML systems can be broadly categorized into three types: Supervised Learning (learning from labeled data), Unsupervised Learning (learning from unlabeled data), and Reinforcement Learning (learning through trial and error with rewards) [7].

2.1.2 Computer Vision

Computer Vision is an interdisciplinary scientific field of AI that deals with how computers can gain high-level understanding from digital images or videos. Its goal is to automate tasks that the human visual system can do [8].

2.1.3 Image Processing

Image Processing refers to the techniques used to perform operations on an image, to get an enhanced image or to extract some useful information from it. In the context of deep learning, this often includes resizing, normalization, and data augmentation [9].

2.1.4 Object Classification

Object Classification, or image classification, is the task of assigning a single label to an entire input image from a predefined set of categories. This task serves as a fundamental benchmark in computer vision [10].

2.1.5 Object Detection

Object Detection is a more complex task that involves identifying and locating objects within an image. The output typically includes a bounding box for each object and a class label [11][12].

2.2 Vision Transformers

The Transformer architecture, originally developed for sequence-to-sequence tasks in Natural Language Processing (NLP), revolutionized deep learning with its reliance on the self-attention mechanism, which allows the model to weigh the importance of different parts of the input relative to others [13].

Vision Transformers (ViT) adapt the Transformer architecture for image recognition tasks. ViT treats an image as a sequence of fixed-size patches, each of which is linearly embedded and then processed by a standard Transformer encoder. This approach demonstrated that a pure Transformer applied directly to image patches could perform comparably or superiorly to Convolutional Neural Networks (CNNs), particularly when trained on very large datasets [14].

2.3 Image Datasets

Several image datasets were utilized in the course of this work. A detailed description of each dataset is provided below.

2.3.1 ImageNet

ImageNet [15] is a large-scale visual database widely used in computer vision research. It is organized according to the WordNet hierarchy and provides annotated, high-resolution images across 1,000 object classes, making it a standard benchmark for large-scale image classification. The dataset contains over 1.2 million training images, 50,000 validation images, and 100,000 test images, covering a broad diversity of real-world objects. Its scale, diversity, and hierarchical structure have established ImageNet as a cornerstone resource for evaluating and advancing machine learning models in image recognition and related tasks.

ImageNetR

ImageNet-R (ImageNet-Renditions) [16] is a dataset derived from ImageNet, consisting of images that depict objects through non-photographic styles such as art, cartoons, graffiti, embroidery, origami, paintings, patterns, sculptures, sketches, tattoos, toys, and video game graphics. It covers 200 ImageNet classes and contains approximately 30,000 images. The dataset was constructed by collecting diverse renditions of ImageNet classes and retaining only those samples that standard models, such as ResNet-50, failed to classify correctly. As a result, ImageNet-R provides a challenging benchmark for evaluating the robustness and generalization of visual recognition models beyond natural images.

ImageNet21k

While ImageNet-1K serves as the primary dataset for pretraining deep learning models for computer vision, ImageNet-21K [17] offers a much broader and more diverse coverage of visual concepts. It is an extended version of the ImageNet dataset and contains over 14 million annotated images spanning approximately 21,000 semantic classes derived from the WordNet hierarchy. Although historically less utilized due to its size, complexity, and accessibility challenges, recent advances in preprocessing, hierarchical structuring, and training schemes have demonstrated its significant value for pretraining. Models pretrained on ImageNet-21K have shown improved performance and generalization across a wide range of computer vision tasks, including when transferred to more compact architectures such as mobile-oriented networks.

ImageNet-21k-P [17] is a cleaned, processed, and optimized subset of the massive ImageNet-21k dataset. The primary goal was to fix the accessibility issues of the original ImageNet-21k (which is huge, unbalanced, and difficult to use) so that more researchers could use it to pretrain powerful models like Vision Transformers (ViT).

2.3.2 CIFAR100

CIFAR-100 [18] is an image classification dataset that extends the structure of CIFAR-10 to a more fine-grained setting. It consists of 60,000 color images of size 32×32 pixels, divided

into 100 classes with 600 images per class. Each class contains 500 training samples and 100 test samples. The classes are grouped into 20 superclasses, providing both a fine label (the specific class) and a coarse label (the superclass). This hierarchical organization allows for evaluating both fine-grained recognition and broader categorization. The dataset includes a wide variety of semantic groups, such as animals (e.g., aquatic mammals, insects, reptiles), objects (e.g., household furniture, vehicles), and natural scenes (e.g., forests, mountains, sea), making it a challenging benchmark for image classification and hierarchical learning tasks.

2.3.3 Caltech256

Caltech-256 [19] is a widely used dataset for object recognition research. It contains 30,607 images distributed across 256 object categories plus an additional "clutter" category for background rejection. Each category includes a minimum of 80 images, providing greater balance compared to earlier datasets such as Caltech-101. The images were collected from the web and manually screened to ensure quality, with improvements introduced to avoid artifacts and increase category diversity. Due to its scale and variability, Caltech-256 serves as a challenging benchmark for evaluating classification performance and the generalization ability of visual recognition models.

2.4 Continual Learning

Continual Learning (CL), also known as lifelong learning, constitutes the core focus of this work. It represents a machine learning paradigm designed to handle the progressive acquisition of knowledge from sequences of tasks or data streams over time. In contrast to conventional training approaches, which rely on a fixed dataset and a single training phase, CL emphasizes the ongoing integration of new information while preserving previously learned knowledge. This property is crucial for real-world scenarios where data distributions are non-stationary and environments evolve continuously. The principal challenge lies in mitigating catastrophic forgetting, a phenomenon in which the assimilation of new knowledge leads to the degradation or loss of prior learning [20].

2.4.1 Elasticity and Plasticity

A central concept in continual learning is the balance between elasticity and plasticity. Plasticity refers to the ability of a model to adapt rapidly to new information, ensuring that it can effectively learn novel tasks. Elasticity, in contrast, emphasizes stability by preserving the parameters and representations that are crucial for retaining old knowledge [4]. Achieving an appropriate equilibrium between these two properties is known as the stability–plasticity dilemma: models that are overly plastic may forget past tasks, whereas models that are overly elastic may struggle to integrate new knowledge. Successful continual learning methods are therefore designed to optimize this trade-off.

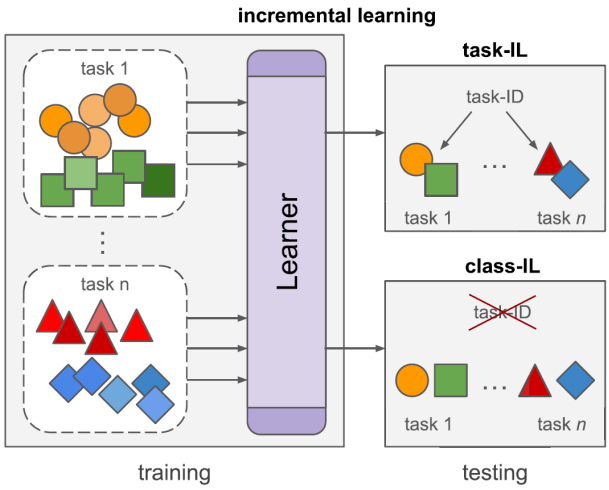


Figure 2.1: Class and Task Incremental Learning [22].

2.4.2 Main Paradigms

As shown in figure 2.1, two principal paradigms define how continual learning problems are formulated. Task-incremental learning (Task-IL) assumes that the task identity is available during inference [21]. In this setting, models can rely on task labels to select the appropriate output head, thereby simplifying the classification problem. Class-incremental learning (Class-IL), on the other hand, does not provide the task identity at test time [22]. The model must classify examples among all the classes encountered so far, which makes the problem significantly more challenging. Class-IL is closer to real-world conditions, as systems are rarely informed of which "task" a new input belongs to during deployment.

2.4.3 Mitigation Strategies

In this work, the discussion of continual learning strategies is organized into three main categories. The first category comprises rehearsal-based methods, which mitigate forgetting by storing or generating representative samples from previous tasks and replaying them during training. The second category encompasses regularization-based methods, which constrain parameter updates to preserve knowledge deemed important for earlier tasks, thereby promoting stability without relying on explicit memory storage. Finally, we include representation-based methods, which explore novel ways of encoding and maintaining information to facilitate continual learning. By focusing on these three categories, the analysis encompasses both classical continual learning approaches and recent innovations, such as prompting mechanisms as an alternative to architectural expansion.

2.4.4 Rehearsal Methods

Rehearsal strategies actively address forgetting by storing and reintroducing past data examples or their generated representations.

- **Experience Replay (ER)** [23] is a rehearsal technique where a small episodic memory buffer is maintained containing a sample of old, representative examples (images and labels). When training on a new task, mini-batches are constructed by mixing data from the new task with a selection of examples from this buffer. This concurrent training helps the model retain a base level of performance on previously learned classes. While it serves as a strong baseline, its main drawbacks are the memory consumption for storing data and potential privacy concerns.
- **Average Gradient Episodic Memory (A-GEM)** [24] improves efficiency by using the buffer to constrain the direction of parameter updates rather than for full data replay. When learning a new task, a reference gradient (g_{ref}) is computed using examples from the small memory buffer. If the new task's current gradient (g) would increase the loss for previous tasks (i.e., $g_{ref} \cdot g < 0$), A-GEM projects the current gradient onto a subspace to ensure the update does not harm the performance on old tasks. This technique effectively balances a minimal memory cost with good stability against catastrophic forgetting.
- **Dark Experience Replay (DER)** [25] enhances ER by incorporating knowledge distillation on the replayed examples. In addition to storing images and labels, DER saves the original model's outputs, or logits (the "dark experience"), for the buffer examples. When training on a new task, the overall loss includes a Distillation Loss that forces the current model's output for the replayed buffer examples to remain close to the saved old logits, thereby maintaining consistency and stability. Dark Experience Replay ++ (DER++) is an evolution of DER that adds a standard Cross-Entropy (CE) Loss using the true labels of the replayed examples alongside the distillation loss.
- **Generative Feature Replay** [26] is a non-data-storing method that tackles both forgetting and class imbalance in CIL. For each task, the model is split into a feature extractor (F_t) and a classifier (H_t). Instead of replaying raw images, a generator (G_{t-1}) is trained to produce synthetic features of previously learned classes. When a new task arrives, the new classifier is trained using real features from the new classes and the generated synthetic features from the old classes, creating a balanced training set. The feature extractor's stability is maintained using a feature distillation loss applied against the old extractor (F_{t-1}). This method leverages the fact that generating features is significantly easier and more privacy-preserving than generating complex images.

2.4.5 Regularization Methods

Regularization methods preserve previously learned information by penalizing modifications to the most important model parameters, which is a technique for ensuring stability.

- **Elastic Weight Consolidation (EWC)** [27] works by identifying parameters critical to prior tasks and restricting their movement during subsequent training. It calculates the Fisher Information Matrix, which estimates the local curvature of the loss function, to determine the "importance" of each parameter for the previous task. A regularization term is then added to the loss function for the new task, acting like a spring that pulls important weights back toward their optimal values (θ_A^*) from the old task, effectively

allowing less important weights to be modified for new learning. EWC ensures that the model can learn Task B without incurring a significant loss on Task.

- **Synaptic Intelligence (SI)** [28] Synaptic Intelligence (SI) provides an alternative, computationally cheaper regularization method compared to EWC. It works by calculating the online importance of each weight by accumulating how much that weight contributed to reducing the loss during the training of the previous task. This importance score (Ω_k^{μ}) is then used in a penalty term in the loss function for the new task, restricting large updates to weights deemed highly important to the old tasks.
- **Learning Without Forgetting (LwF)** [29] is a distillation-based method that regularizes the model's output rather than its weights. When learning a new task, the model trained on the previous task is kept as a teacher model. The new model (student) is trained not only with the CE Loss for the new task but also with a Distillation Loss that forces the student's output for the new data to match the teacher's soft targets (logits) for the old classes.
- **Adaptive Plasticity Improvement (API)** [4] employs Dual Gradient Projection Memory (Dual-GPM) to enforce an update direction that is orthogonal to the important subspaces of previous tasks, preventing the new gradient from harming old knowledge. API also introduces a metric called the Average Gradient Retention Ratio (AGRR) to measure the model's plasticity. If the AGRR indicates insufficient learning capacity (because most of the new task's gradient is being projected out), API will adaptively expand the network by adding new dimensions/units to the layers, creating necessary space for the new knowledge.
- **Inference Free Low-Rank Adaptation (InFLoRA)** applies a parameter-efficient technique, adding new, low-rank adaptors (A_t, B_t) to the attention layers of a frozen backbone for each new task. It uses Dual-GPM to design the dimensionality reduction matrix B_t such that the update space is interference-free, meaning it is orthogonal to the gradients of old tasks, ensuring the model learns the new task without forgetting the old. The backbone and old adaptors remain frozen, and only the new adaptor A_t is fine-tuned. The final weight for task t is an accumulation of the frozen backbone and all trained adaptors ($W_t = W + \sum_{j=1}^{t} A_j B_j$). **Distributed Online LoRA for Federated INcremental learning method (DOLFIN)** [30] extends InFLORA to Federated Learning, distributing the Dual-GPM process across multiple clients. Each client locally calculates its orthogonal basis and contributes to the global adaptor, which is highly communication-efficient and preserves data privacy

2.4.6 Representation Methods

These methods encode new knowledge by working with the model's existing feature space or input embedding, often by keeping the model backbone frozen and inserting small, trainable vectors to give more context or choosing an specific representations for the image classes and information.

- **Learning to Prompt (L2P)** [31] works by keeping the pre-trained model frozen and training a small prompt pool which acts as a memory of instructions. For a new input, a

query function is used to select the most relevant prompts from the pool. These selected prompt vectors are then prepended to the input embedding (or tokens), guiding the frozen backbone to the correct output without changing its original weights.Learning to Prompt (L2P) works by keeping the pre-trained model frozen and training a small prompt pool which acts as a memory of instructions. For a new input, a query function is used to select the most relevant prompts from the pool. These selected prompt vectors are then prepended to the input embedding (or tokens), guiding the frozen backbone to the correct output without changing its original weights.

- **Dual Prompt** [32] refines L2P by separating the learned representation into two distinct prompt sets. It uses fixed G-prompts (General) which are always attached and continuously updated to capture general knowledge (stability). Simultaneously, it uses task-specific E-prompts (Expert) which are selected via a query and updated only when matched, capturing new specific knowledge (plasticity). Both sets of prompts are attached to multiple Multi-head Self-Attention layers.
- **CODA-Prompt** [33] introduces a fully differentiable method for creating the instruction representation. Instead of discretely selecting a prompt, the final prompt is constructed as a weighted sum (Σ) of multiple learned prompt components. The weights (α) are dynamically calculated via an attention mechanism based on the input query-key similarity. This allows the entire prompt-generation process to be optimized end-to-end, improving plasticity and allowing the set of prompt components to expand over.

In summary, as illustrated in Figure 2.2, the conceptual framework of continual learning can be understood through four interrelated components. First, continual learning involves adapting to incremental tasks under dynamic data distributions, reflecting the non-stationary characteristics of real-world environments. Second, an effective continual learning system must strike an appropriate balance between stability and plasticity: stability ensures the retention of previously acquired knowledge, whereas plasticity enables the integration of novel information. Moreover, the model should generalize effectively to both intra-task variations, which represent differences within a single task, and inter-task variations, which capture distributional shifts across tasks. Third, to achieve these objectives, a variety of methodological approaches have been proposed—each addressing different facets of the learning process. Unlike those presented in [20], the state-of-the-art methods analysed in this work are categorized into rehearsal, regularization, and representation-based approaches. Finally, continual learning techniques must also be tailored to practical scenarios, where factors such as task specificity, scenario complexity, and computational limitations further influence the design and evaluation of effective solutions [20].

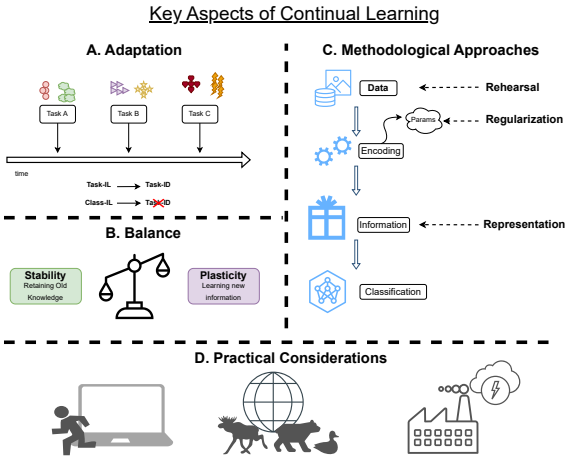


Figure 2.2: Key aspects of Continual Learning.

Chapter 3

Methodology

Our approach enables an image recognition system to continually learn new object classes over time without forgetting previously learned ones. This catastrophic forgetting occurs when models adapt to new information but demonstrate reduced or poor performance on earlier tasks they had once mastered. Addressing this loss of performance is essential for building adaptive and reliable autonomous systems and is achieved by pairing an unsupervised pretrained feature extractor with independent Gaussian models that capture each class's representation. We begin by describing how the features are extracted using a foundation model in section 3.1 In section 3.2 the way the Gaussian representations are generated is described. Finally, in section 3.3, we explain the inference step and the incremental addition of new classes by presenting the experimental methodology.

3.1 Backbone description

The primary requirement for a feature extraction model to be used as a backbone is that it must be pretrained — before the recognition system is initialized or created — and the features obtained must be sufficiently general to characterize all possible classes of objects to be seen, not just the ones known at training time.

3.1.1 Training the backbone for image classification

To validate the proposed concept, we employed a ViT-based foundation model [14] as the feature extractor, utilizing ImageNet [15] for training. Specifically, the model was pre-trained on ImageNet-21k and subsequently fine-tuned on ImageNet-1k for image classification. In this work, the classification head was discarded; the remaining feature extraction layers were preserved and frozen to serve as a static backbone.

3.1.2 Using a modular backbone for continual learning

Implementing a modular backbone within a continual learning framework mitigates catastrophic forgetting. By relying exclusively on generalized feature extraction and preventing updates to the backbone weights, the system maintains a stable representation across incremental tasks.

3.2 Gaussian representations using extracted embeddings

Assuming each object class is single-modal and can therefore be represented by a Gaussian distribution, a single Gaussian representation is used for each class. Two variants are implemented: a separable Gaussian model, which assumes diagonal covariance matrices (e.g., the naive Bayes assumption), and a non-separable Gaussian model, which uses full covariance matrices.

3.2.1 Separable Gaussian representation

In the separable Gaussian model, it is required to store the following values for each object class: the sum of each component of the feature vector, the sum of the squares of each component of the feature vector, and the number of samples per class. Using those stored statics, it is possible to to calculate — and incrementally update as new data becomes available — each class separable Gaussian representation as shown in algorithm 1.

3.2.2 Non-Separable Gaussian representation

In the non-separable Gaussian model, instead of storing scalar values, the sum of the feature vectors x_i and the sum of outer products of the feature vectors $x_i x_i^T$ are stored. In the same way as the separable case, the previous described statics allow the incremental update of each class representation as shown in algorithm 2.

3.2.3 Two-mode Gaussian representationg

Since object classes do not necessarily have single-modal distributions, a two-mode Gaussian Mixture Model (GMM) can be used to represent each class. In this can be done by using separable or non-separable covariance matrices.

Algorithm 1 Separable Gaussian Modeling

if class statistics exist **then**
    Load `sums`, `sums2`, and `class_counts`
else
    Initialize `sums`, `sums2`, and `class_counts` to zero for each class
end if
for each feature vector $\mathbf{x}$ with class label k **do**
    Update $\texttt{sums}_k \leftarrow \texttt{sums}_k + \mathbf{x}$
    Update $\texttt{sums2}_k \leftarrow \texttt{sums2}_k + \mathbf{x}^2$
    Update $\texttt{class_counts}_k \leftarrow \texttt{class_counts}_k + 1$
end for
for each class k **do**
    Compute mean: $\mu_k = \frac{\texttt{sums}_k}{\texttt{class_counts}_k}$
    Compute variance: $\sigma_k^2 = \frac{\texttt{sums2}_k}{\texttt{class_counts}_k} - \mu_k^2$
    Add stability term: $\text{cov}_k = \sigma_k^2 + \varepsilon$
    Compute log-normalization constant: $\ln A_k = -\frac{d}{2}\log(\pi) - \frac{1}{2}\sum_{j=1}^{d}\log(\sigma_{k,j}^2)$
end for
function CLASSIFY($\mathbf{x}$)
    for each class k **do**
        Compute Mahalanobis distance: $D_k = \sum_{j=1}^{d}\frac{(x_j-\mu_{k,j})^2}{\sigma_{k,j}^2}$
        Compute log-likelihood: $\log p_k(\mathbf{x}) = \ln A_k - \frac{1}{2}D_k$
    end for
    return class k with maximum $\log p_k(\mathbf{x})$
end function

Algorithm 2 Non-Separable Gaussian Modeling

if class statistics exist **then**
    Load $\texttt{sums}$, $\texttt{outer_sums}$, and $\texttt{class_counts}$
else
    Initialize $\texttt{sums}$, $\texttt{outer_sums}$, and $\texttt{class_counts}$ to zero for each class
end if
for each feature vector $\mathbf{x}$ with class label k **do**
    Update $\texttt{sums}_k \leftarrow \texttt{sums}_k + \mathbf{x}$
    Update $\texttt{outer_sums}_k \leftarrow \texttt{outer_sums}_k + \mathbf{x}\mathbf{x}^\top$
    Update $\texttt{class_counts}_k \leftarrow \texttt{class_counts}_k + 1$
end for
for each class k **do**
    Compute mean: $\mu_k = \frac{\texttt{sums}_k}{\texttt{class_counts}_k}$
    Compute covariance: $\Sigma_k = \frac{\texttt{outer_sums}_k}{\texttt{class_counts}_k} - \mu_k \mu_k^\top$
    Add regularization: $\Sigma_k \leftarrow \Sigma_k + \varepsilon I$
    if Σ_k is not positive semi-definite **then**
        Perform eigendecomposition: $\Sigma_k = P \Lambda P^\top$
        Replace $\Lambda \leftarrow \max(\Lambda, \varepsilon)$
        Reconstruct covariance: $\Sigma_k \leftarrow P \Lambda P^\top$
    end if
    Compute log-normalization: $\log A_k = -\frac{1}{2}\left(d \log(2\pi) + \log |\Sigma_k|\right)$
end for
function CLASSIFY($\mathbf{x}$)
    for each class k **do**
        Compute Mahalanobis distance: $D_k = (\mathbf{x} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x} - \mu_k)$
        Compute log-likelihood: $\log p_k(\mathbf{x}) = \log A_k - \frac{1}{2} D_k$
    end for
    return class k with maximum $\log p_k(\mathbf{x})$
end function

3.2.4 Up-to-four mode Gaussian representation with modality testing

Because the number of modes required to represent an object class is variable, the algorithm in Algorithm 3 is used to determine the necessary number of modes — up to four modes in this case — and, consequently, the number of Gaussians.

Algorithm 3 GMM Training

Input: Features $\mathcal{X}_k$ per class k.
Output: Number of gaussians M_k per class k
GMM means and covariances $\{(\boldsymbol{\mu}_{k,m}, \boldsymbol{\Sigma}_{k,m})\}$ per class k
function TRAINGMM($\mathcal{X}_k$)
  for $j = 2$ to 4 **do**
    Generate j clusters from $\mathcal{X}_k$ using k-means
    Compute MCP_j (max cluster percentage)
    Compute CH_j (Calinski-Harabasz Score)
    Compute $S_j \leftarrow \text{MCP}_j - \text{CH}_j/2$ (general score)
  end for
  $M_k \leftarrow \arg\max_j S_j$ **if** $\max_j S_j > score_{th}$ **else** 1
  Compute Gaussian Mixture parameters $\{(\boldsymbol{\mu}_{k,m}, \boldsymbol{\Sigma}_{k,m})\}$ considering M_k gaussians
  return M_k, $\{(\boldsymbol{\mu}_{k,m}, \boldsymbol{\Sigma}_{k,m})\}$
end function

3.3 Experimental Methodology

To evaluate the efficacy of our approach in a Class-Incremental Learning (CIL) setting, we conduct experiments on two standard benchmarks: Split CIFAR-100 and Split ImageNet-R.

3.3.1 Datasets and Experimental Protocol

We follow the standard Continual Learning protocol where the model learns a sequence of disjoint tasks. The specific configurations for the datasets are as follows:

- Split CIFAR-100: This dataset is a widely utilized benchmark in the continual learning literature. It partitions the original CIFAR-100 [18] dataset into 10 disjoint tasks, with each task containing 10 distinct classes.
- Split ImageNet-R: This benchmark is constructed from ImageNet-R [16], a dataset containing renditions of ImageNet classes (e.g., art, cartoons, sketches). We randomly divide the 200 classes into 10 sequential tasks, with 20 classes per task.

3.3.2 Evaluation Metrics

To quantify performance, and inspired by the Learn to Prompt work [31] we utilize two standard metrics derived from the test accuracy scores: the average accuracy and the average forgetting.

Average Accuracy (A_t), presented on equation (3.1), measures overall performance after task t. It is the mean test accuracy across all tasks learned up to that point. The final A_{10} (after all 10 tasks) is the number reported in benchmark tables of the L2P [31] work.

$$A_t = \frac{1}{t} \sum_{\tau=1}^{t} S_{t,\tau} \tag{3.1}$$

Forgetting (F_t), presented on equation (3.2), quantifies how much the model has forgotten earlier tasks. It computes, for each past task, how much accuracy has dropped compared to its best historical score, and averages those drops.

$$F_t = \frac{1}{t-1} \sum_{\tau=1}^{t-1} \left(\max_{k \leq t-1} S_{k,\tau} - S_{t,\tau} \right) \tag{3.2}$$

Chapter 4

Results

Our approach uses a pre-trained foundation model to extract features of images and generate Gaussian representations per class where, during the inference step, the classifier will predict the class that hast the highest likelihood among the existing Gaussians. Several Gaussian representations were developed: separable, non-separable, and Gaussian-mixture models using both separable and non-separable mixtures.

4.1 Baselines Comparison

We selected prompt-based methods, specifically L2P [31], Dual Prompt [32], and CODA-Prompt [33], as our baselines. These methods utilize a ViT-B/16 [14] as the backbone. For our experiments, we evaluated our model using two distinct backbones: the standard ViT-B/16 and a ViT-M/16 [34]. Both architectures were pre-trained on ImageNet-21k and subsequently fine-tuned on ImageNet-1k. The performance of these backbones on ImageNet-1k, using the original FFN classification head, is presented in Table 4.1.

Table 4.1: ImageNet Accuracy of ViT Backbones

Backbone	**Imagenet Accuracy**
ViT-B/16	81.07%
ViT-m/16	87.44%

Table 4.2 presents the experimental results on the CIFAR-100 and ImageNet-R datasets. While CIFAR-100 is class-balanced, ImageNet-R exhibits significant imbalance, with sample counts varying widely across classes. To address this, we applied a balancing strategy in select experiments, denoted by the suffix "-B". In these cases, synthetic samples were generated for under-represented classes by sampling from a Gaussian distribution defined by the class mean and covariance. Regarding the GMM-based methods, the suffix "-S" indicates the use of separable covariance, while "-N" denotes non-separable covariance. If no covariance suffix is present, non-separable covariance is implied.

Table 4.2: Comparison of Method Performance on Split CIFAR 100 and Split Imagenet R

Method	Split CIFAR 100		Split Imagenet R	
	Avg. Acc	**Forgetting**	**Avg. Acc**	**Forgetting**
L2P	83.86	7.35	61.57	9.73
DualPrompt	86.51	5.16	68.13	4.68
CODAPrompt	86.25	0.74	75.45	1.64
Separable ViTb	78.63	17.52	52.56	4.89
Separable-B ViTb	–	–	52.79	5.33
Non-Separable ViTb	82.36	20.73	41.72	15.41
Non-Separable-B ViTb	–	–	46.46	12.19
GMM-2-NS ViTb	75.34	20.73	42.56	17.71
GMM-2-S ViTb	53.53	17.85	53.53	1.18
GMM-up-to-4-NS ViTb	82.34	14.07	44.43	17.85
GMM-up-to-4-S ViTb	80.40	15.55	50.51	3.48
Separable ViTm	86.28	12.02	56.16	16.55
Separable-B ViTm	–	–	56.34	18.89
Non-Separable ViTm	88.12	10.20	56.66	18.89
Non-Separable-B ViTm	–	–	56.66	18.89
GMM-2-NS ViTm	83.37	5.25	59.21	8.48
GMM-2-S ViTm	83.51	3.97	70.37	3.14
GMM-up-to-4 ViTm	85.71	3.64	55.01	19.87
GMM-up-to-4-S ViTm	83.60	4.69	68.32	7.98

4.2 Dimensionality Reduction

To reduce the dimensionality of the 768-dimensional ViT-B/16 embeddings, we trained a *foundation PCA* — defined here as a PCA trained with sufficient samples to generalize across diverse feature distributions. Table 4.3 presents the results for both separable and non-separable methods across different PCA configurations. Methods annotated with the suffix "-F" utilize a PCA trained on the full ImageNet-21k dataset. In the absence of this suffix, the PCA was trained on the ImageNet-21k-P subset [17].

4.3 Top Accuracies Results

Finally, Tables 4.4 and 4.5 present the results of various experiments using both our separable and non-separable methods over CIFAR100 and ImagenetR, specifically including top-3 and top-5 accuracy metrics. These additional metrics allow for a more in-depth analysis of our model's performance characteristics.

Table 4.3: Performance comparison of PCA-based ViT methods against baselines

Method	**Split CIFAR 100**		**Split Imagenet R**	
	Avg. Acc	**Forgetting**	**Avg. Acc**	**Forgetting**
L2P	83.86	7.35	61.57	9.73
DualPrompt	86.51	5.16	68.13	4.68
CODAPrompt	86.25	0.74	75.45	1.64
Separable ViTb	78.63	17.52	52.56	4.89
Non-Separable ViTb	82.36	20.73	41.72	15.41
Separable ViTb PCA64	66.04	24.96	25.99	17.22
Separable ViTb PCA128	72.58	21.69	34.84	12.57
Separable ViTb PCA256	76.76	18.49	46.25	9.05
Separable ViTb PCA768	79.59	16.90	56.25	2.21
Separable ViTb PCA64-F	66.07	25.15	–	–
Separable ViTb PCA128-F	72.68	21.80	–	–
Separable ViTb PCA256-F	76.62	19.09	–	–
Separable ViTb PCA768-F	79.52	16.87	–	–
Non-Separable ViTb PCA64	73.37	18.48	31.88	16.16
Non-Separable ViTb PCA128	75.42	18.20	21.47	15.52
Non-Separable ViTb PCA256	73.12	18.65	31.04	14.36
Non-Separable ViTb PCA768	75.66	21.49	46.83	15.12

Table 4.4: Comparison of Top-1, Top-3, and Top-5 Accuracy on Split CIFAR 100

Method	**Split CIFAR 100**					
	Top 1		**Top 3**		**Top 5**	
	Acc	**Forg**	**Acc**	**Forg**	**Acc**	**Forg**
Separable ViT	78.63	17.52	91.51	8.88	94.41	6.10
Non-Separable ViT	82.36	14.05	93.81	6.43	96.28	4.02
Separable ViT PCA768	79.59	16.90	91.70	8.66	94.38	6.02
Separable ViT PCA256	76.76	18.49	90.61	9.76	93.73	6.95
Separable ViT PCA128	72.58	21.69	88.59	11.46	92.56	7.72
Separable ViT PCA64	66.04	24.96	84.83	15.18	90.38	10.25
Non-Separable ViT PCA768	75.66	21.49	93.82	6.42	96.24	3.95
Non-Separable ViT PCA256	73.12	18.65	87.40	11.22	91.39	8.90
Non-Separable ViT PCA128	75.42	18.20	89.32	9.65	92.79	7.45
Non-Separable ViT PCA64	73.37	18.48	88.48	10.80	92.31	7.88

Table 4.5: Comparison of Top-1, Top-3, and Top-5 Accuracy on Split Imagenet-R

Method	**Split Imagenet-R**					
	Top 1		**Top 3**		**Top 5**	
	Acc	**Forg**	**Acc**	**Forg**	**Acc**	**Forg**
Separable	52.56	4.89	65.92	9.80	71.20	12.95
Non-Separable	41.72	15.41	54.85	21.75	61.67	24.04

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